

PARTH SHARMA

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SUMMARY

GenAI / LLM Engineer (B.Tech, Artificial Intelligence & Data Science) with hands-on experience in **Python, LLMs, Retrieval-Augmented Generation (RAG), APIs, and FastAPI**. Combines GenAI skills with a solid data foundation: SQL, ETL pipelines, data cleaning, and validation on 500K+ records, which matters when building retrieval pipelines that depend on clean, well-structured data. Experienced in ML fundamentals and feature engineering, and in delivering insights to non-technical stakeholders.

EXPERIENCE

GenAI / LLM Engineer Intern — Decode Data Academy July 2025 – Present

- Cleaned, standardized, and structured 10,000+ records across 5+ branches and 2 colleges with **SQL/MySQL pipelines**, eliminating 95%+ of data errors, the kind of clean, well-structured data that retrieval and LLM applications depend on.
- Designed a normalized relational schema and 10–15 SQL views so data is easy to query and consume by downstream applications.
- Automated repeatable ingestion and transformation workflows with a **Pentaho Spoon** ETL layer, improving processing efficiency by 20–30%.
- Implemented validation and business-rule checks across 1,000+ records to catch inconsistent or malformed entries before downstream use.
- Translated requirements from a 6-member analytics team into usable outputs, presenting insights through 6 Power BI dashboards to non-technical users.

PROJECTS

Insurance Incentive Optimization ([GitHub](#), [Live](#)) Python, FastAPI, Scikit-learn

- Built and deployed an end-to-end ML service that predicts **policy renewal probability** on 67,875 records and recommends a revenue-maximizing agent incentive per policy.
- Designed a **FastAPI /predict** REST endpoint (Pydantic validation, CORS, batch JSON input/output) on Render, consumed by a live frontend on Vercel, the same API pattern used to serve LLM/RAG backends.
- Built a scikit-learn Pipeline (imputation, one-hot/target encoding, scaling, engineered features) and persisted it with joblib/pickle for low-latency inference.

Loan Default Risk Prediction ([GitHub](#)) Python, Pandas, Scikit-learn

- Built an end-to-end **ML pipeline** on a 55K-row, 122-column loan dataset: missing-value and outlier handling, encoding, and model-ready data preparation.
- Engineered **10+ predictive features** (income segmentation, credit ratio, EMI burden, employment duration) to improve customer risk profiling.
- Trained and compared **Logistic Regression, Decision Tree, and Gradient Boosting** classifiers to predict loan default risk; evaluated with standard classification metrics.

- Flight Data Engineering Pipeline (GitHub)Pentaho, MySQL, Power BI
- Built an end-to-end **data pipeline** (Pentaho Spoon, MySQL) processing 445K+ flight records into an analytics-ready fact table (455K rows → 6.5K, ~98% reduction), the kind of feature-ready data layer ML workloads depend on.

– Modeled star-schema tables and SQL views feeding 4 Power BI dashboards on airline delays, airport performance, and carrier efficiency.

EDUCATION

- 2022 - 2026Bachelor of Technology (Artificial Intelligence and Data Science) at **Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh, India**
- 2020 - 2021Higher Secondary Education (Class XII) at **Manikala Nahar Higher Secondary School**
- 2018 - 2019Secondary Education (Class X) at **Shri Maniklal Nahar International School**

TECHNICAL SKILLS

- Programming:

GenAI & LLMs:

APIs & Backend:

Machine Learning:

Data Engineering:

Libraries:

Databases:

Tools:

Coursework:
- Python, SQL

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Generative AI

FastAPI, REST APIs, API Integration

Feature Engineering, Classification (Logistic Regression, Decision Trees, Gradient Boosting), Model Evaluation

ETL Pipelines, Data Cleaning, Data Validation, Data Transformation

Pandas, NumPy, Scikit-learn, XGBoost

MySQL, Relational Databases

Git, Jupyter Notebook, Power BI

Machine Learning, DBMS, Statistics, Data Structures and Algorithms

CERTIFICATIONS

- AWS Academy Data Engineering Badge

– Wipro TalentNext Database Solutions Expert

– HackerRank SQL – 5-Star (Gold) Badge
- Dec 2025

Oct 2024